

k-means clustering is to minimise the distance of data points from the centroid of this cluster and maximise the distance from other cluster centroids. So in this step, we have to find the closest centroid to each data point.

5. *Re-compute cluster centroids*: The new centroids will be calculated as the mean of the points that belong to the centroid of the previous step.

In a nutshell, the steps involved in the k-means algorithm are:

1. Randomly place 'n' centroids, one for each cluster.
2. Calculate the distance of each data point from each centroid.
3. Assign each data point to its closest centroid, creating a cluster.
4. Recalculate the position of centroids.
5. Since k-means is an iterative algorithm, repeat Steps 2 to 4 until it results in the clusters with minimum error or the centroids no longer move.

Implementing the k-means algorithm

Initially, import the following libraries — Pandas, Numpy and Matplotlib. Next, generate some random data in a two-dimensional space. Here, a 100 data points have been generated and divided into two groups of 50 points each.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
X= -2 * np.random.rand(100,2)
X1 = 1 + 2 * np.random.rand(50,2)
```

```
In [5]: from sklearn.cluster import KMeans
        Kmean = KMeans(n_clusters=2)
        Kmean.fit(X)

Out[5]: KMeans(algorithm='auto', copy_x=True, init='k-means++', max_iter=300,
              n_clusters=2, n_init=10, n_jobs=None, precompute_distances='auto',
              random_state=None, tol=0.0001, verbose=0)

In [6]: Kmean.cluster_centers_

Out[6]: array([[ -1.13660621, -1.0979858 ],
               [  2.06037688,  1.8179407 ]])
```

Figure 4: Centroid values

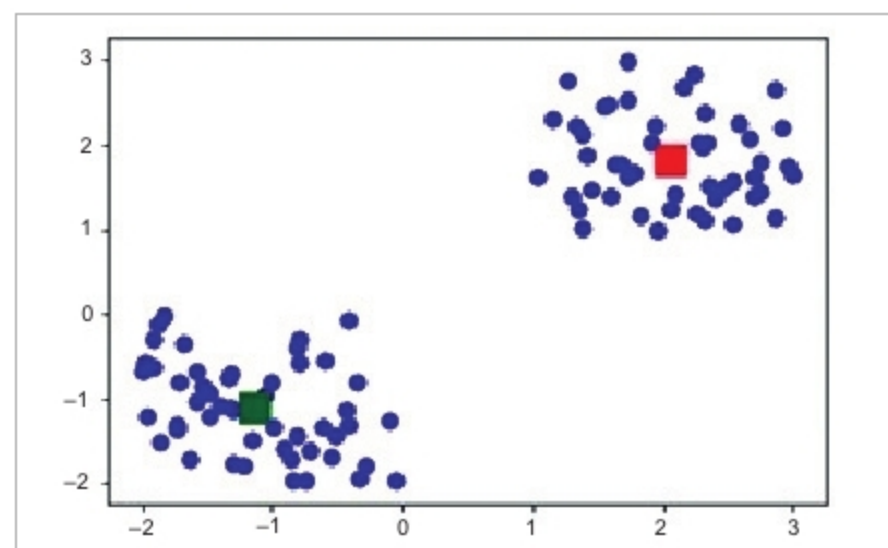


Figure 5: Displaying the cluster centroids (green and red colour)

```
X[50:100, :] = X1
plt.scatter(X[:, 0], X[:, 1], s = 50, c = 'b')
plt.show()
#In this case, we arbitrarily gave k (no of clusters) a value
of two.
#Next, we need to find the centre of the clusters, centroid by
using Kmean.cluster_centres

from sklearn.cluster import KMeans
Kmean = KMeans(n_clusters=2)
Kmean.fit(X)
Kmean.cluster_centers_
#Hence we get the value of centroids, -1.13660621, -1.0979858
and 2.06037688, 1.8179407
plt.scatter(X[:, 0], X[:, 1], s=50, c='b')
plt.scatter(-0.94665068, -0.97138368, s=200, c='y',
marker='s')
plt.scatter(2.01559419, 2.02597093, s=200, c='r', marker='s')
plt.show()
```

The k-means algorithm identifies 'k' number of centroids and then allocates every data point to the nearest cluster, while keeping the centroids as small as possible. The 'means' in k-means refers to averaging of the data, i.e., finding the centroid.

The correct choice of 'k' is often ambiguous because it's very dependent on the shape and scale of the distribution of points in a data set.

The elbow method is used for determining the correct number of clusters in a data set. It works by plotting the ascending values of 'k' versus the total error obtained when using that 'k'. This means increasing 'k' will always decrease the error. So, the value of the metric as a function of 'k' is plotted and the elbow point is determined where the rate at which the error decreases shifts sharply. It is the right 'k' for clustering.

So, k-means is partition based clustering, which is relatively efficient on medium and large sized data sets. It produces sphere-like clusters because the clusters are shaped around the centroids. Its drawback is that we need to pre-specify the number of clusters, which is not going to be an easy task.

Density based clustering algorithms

Most traditional clustering techniques such as k-means, hierarchical and fuzzy clustering can be used to group data in an unsupervised way. However, when applied to tasks with arbitrary shaped clusters or to clusters within clusters, traditional techniques might not be able to achieve good results, i.e., elements in the same cluster might not share enough similarity or the performance may be poor.

In contrast, density based clustering locates regions of high density that are separated from one another by regions of low density. Density in this context is defined as the number of points within a specified radius.